

RL电路的零输入问题

$$i_L(0+) = i_L(0-) = \frac{U_s}{R_1 + R} = I_0$$

$$R i_L + L \frac{di_L}{dt} = 0 \quad t \geq 0$$

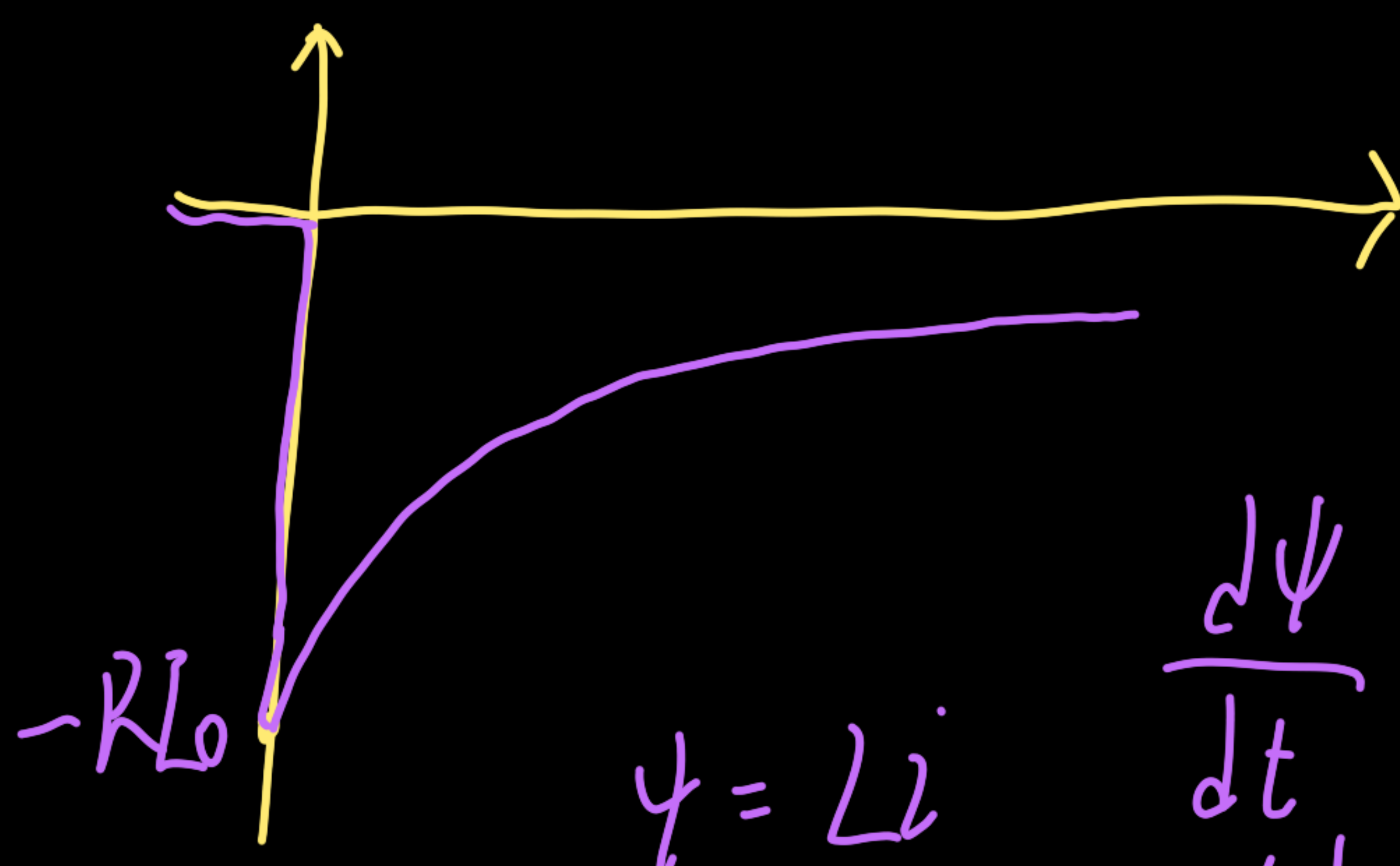
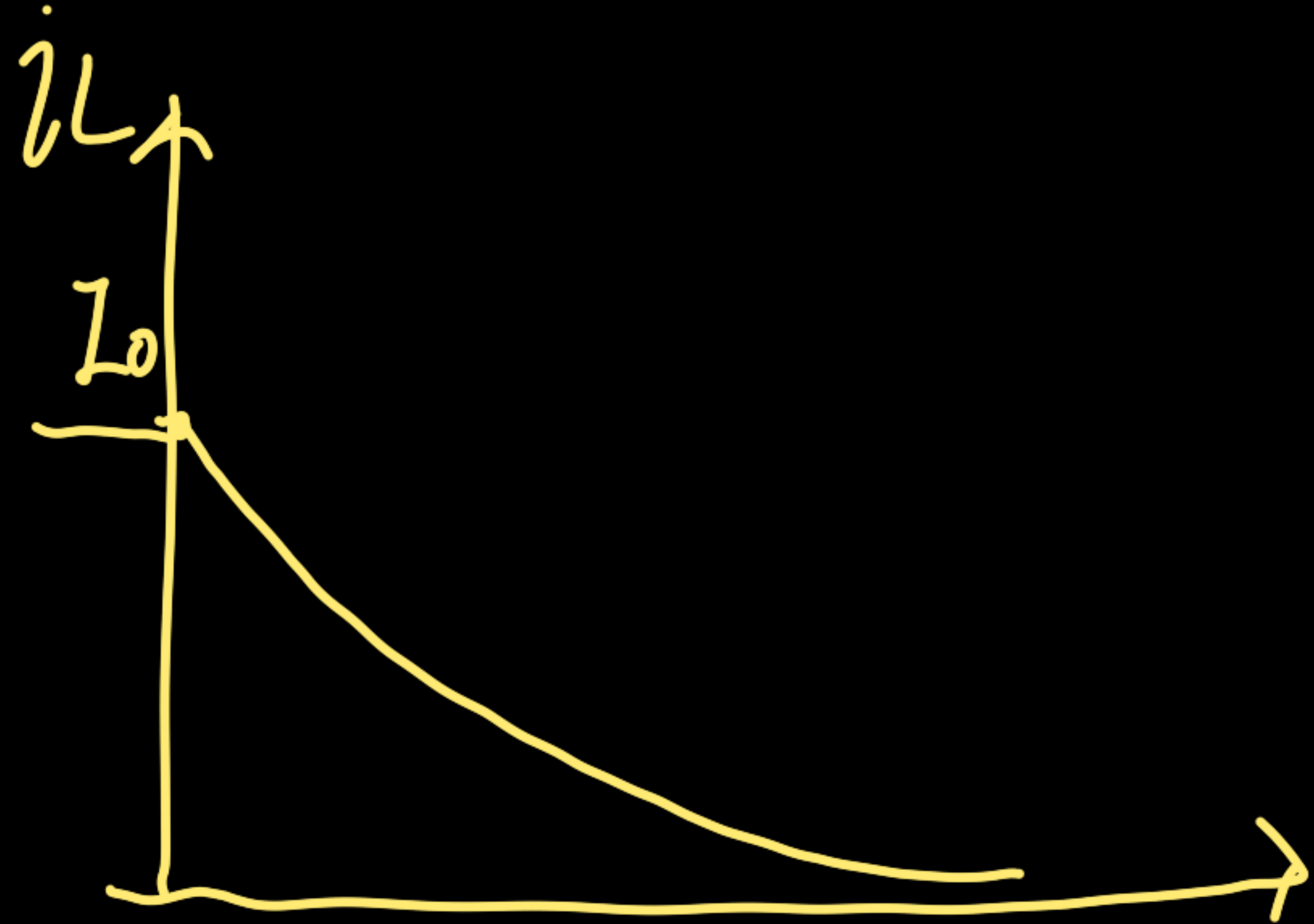
$$R + Lp = 0 \quad p = -\frac{R}{L}$$

$$i_L(t) = A e^{pt} = A e^{-\frac{R}{L}t}$$

$$i_L(0) = A e^0 = I_0 \quad A = I_0$$

$$i_L(t) = I_0 e^{pt} = I_0 e^{-\frac{R}{L}t} \quad t \geq 0$$

$$U_L(t) = L \frac{di_L}{dt} = L \cdot \left(-\frac{R}{L} I_0 e^{-\frac{R}{L}t}\right) = -R I_0 e^{-\frac{R}{L}t}$$



$$\tau = -\frac{L}{R} \quad [L] = \left[\frac{\text{亨}}{\text{欧}}\right] = \left[\frac{\text{韦}}{\text{欧·安}}\right] = \left[\frac{\text{伏·秒}}{\text{欧·安}}\right] = [\text{秒}]$$

③ L 越大, R 小, $W = \frac{1}{2} L i^2$ 起始能量大, $P = R i^2$ 放电过程消耗小, 放电时间长, 放电慢, τ 大

$$i_L(0+) = i_L(0-) = \frac{24}{2+4+3//6} \times \frac{2}{3} = 2\text{A}$$

$$3+6//6 = 6\Omega \quad \tau = \frac{L}{R} = 1\text{s} \quad i_L = 2e^{-t}\text{A}$$

$$U_L = i_L R = -12e^{-t}\text{V} \quad t \geq 0$$

$$U_{12} = 24\text{V} + 4 \times \frac{U_L}{2} = 24 + 4e^{-t}\text{V}$$

